A SUMMARIZED REPORT ON NINE MAJOR NETWORK SIMULATION AND EMULATION TOOLS

(GROUP 4)

NETWORK SIMULATOR

A **network simulator** is a software program that can predict the performance of a computer network or a wireless communication network. Since communication networks have become too complex for traditional analytical methods to provide an accurate understanding of system behavior, network simulators are used. In simulators, the computer network is modeled with devices, links, applications, etc., and the network performance is reported. Simulators come with support for the most popular technologies and networks in use today such as 5G/6G, Satellite Networks, internet of things (IoT), wireless LANs, mobile ad hoc networks, wireless sensor networks, vehicular ad hoc networks and many more.

This report discusses 9 major Network Simulation Tools.

1. OMNeT++

OMNeT++ (Objective Modular Network Testbed in C++)

OMNeT++ is a widely used, component-based, C++ discrete event simulation framework that provides an ecosystem for building network simulators and models of distributed systems. It is particularly popular in academic and research circles due to its modular design and extensive tooling.

Aspect Summary

Core Philosophy

Provides a generic simulation platform and infrastructure, not a simulator of a specific concrete thing. Models are built as a hierarchy of reusable modules.

Primary Language

C++, with a topology description language (NED) and INI files for configuration.

Key Features

- Hierarchical, component-based architecture for high reusability.

- Integrated Development Environment (IDE) with a graphical editor for network topologies.

- Rich GUI-based execution environment (Qtenv/Tkenv) for animation, debugging, and tracing.

- Rule-based result analysis and robust statistical collection tools.

- Support for parallel and distributed simulation (PDES) using MPI for large-scale models.

Use Cases

- Modeling wired and wireless communication networks (via the INET Framework model).

- Sensor networks, mobile ad-hoc networks, peer-to-peer, optical networks, and queuing systems.

- Performance evaluation of complex software systems and hardware architectures.

Strengths

- High modularity and reusability of components.

- Excellent visualization, debugging, and analysis tools reduce development time.

- Clear separation of model logic (C++), topology (NED), and experimental parameters (INI files).

- Ability to use real protocol implementations (e.g., portions of the FreeBSD TCP/IP stack) within simulations.

Weaknesses

- The core is a framework, so users must rely on third-party models (like INET) for specific protocols, which have their own release cycles.

- The learning curve for the C++, NED, and INI file interaction can be steep for new users.

2. NS-3 (Network Simulator 3)

NS-3 is an open-source, discrete-event network simulator designed for networking research and education. It emphasizes realism, integration with real systems and testbeds, and community-driven development, providing a modern alternative to its predecessor, NS-2.

Aspect Summary

Core Philosophy

Develop a free and open-source simulation environment suitable for modern networking research with an emphasis on code realism and integration.

Primary Language

C++ with optional Python bindings for scripting; the entire kernel is in C++.

Key Features

- Integration with real-world code through frameworks like Direct Code Execution (DCE), allowing the use of real Linux kernel networking stacks.

- High fidelity and realism in models, particularly for wireless systems like Wi-Fi and LTE.

- Flexible tracing and statistics frameworks that can generate pcap files for use with external analysis tools (like Wireshark).

- Support for "simulation-in-the-loop" use cases, allowing interaction with real network devices and virtual machines.

Use Cases

- Studying Internet protocols (TCP/IP) and large-scale systems.

- Research on wireless networks, including Wi-Fi, WiMAX, and LTE.

- Mobile ad hoc routing protocol performance analysis.

- Modeling non-Internet based systems.

Strengths

- Strong emphasis on realism and the ability to integrate real application/kernel code.

- Active development and maintenance by a global volunteer community.

- Excellent scalability for large networks (hundreds of millions of nodes have been simulated).

- Open-source license (GNU GPLv2) encourages contribution and peer review.

Weaknesses

- Does not include a single integrated GUI; users rely on external animators (like NetAnim or PyViz) and command-line tools.

- Primarily used on Linux/macOS systems; native Windows support is not a primary focus, often requiring the Windows Subsystem for Linux (WSL)

3. QualNet

QualNet is a commercial network simulation tool owned by Keysight Technologies (formerly Scalable Network Technologies).

QualNet uses parallel discrete event simulation technology to model various layers of the network stack including the physical layer of wireless radios. QualNet simulations may be customized to include terrain data, antenna patterns, experimental or specialized network protocols (ex mesh networking), and mobility (nodes move during the simulation). Alternatives to QualNet include NS-2 and OPNET. QualNet was used since the early 2000s to research and develop wireless communication technologies.

In addition, QualNet provides a **network digital twin**, allowing users to model devices, protocol layers, radios, antennas, and the operating environment. It supports heterogeneous networks including wired, wireless, satellite, and underwater systems with thousands of nodes. Using **Parallel Discrete Event Simulation (PDES)** algorithms, QualNet runs high-fidelity simulations faster than real-time, enabling rapid "what-if" analyses.

The platform features an intuitive **GUI** with **Design Mode** for building network topologies, **Visualize Mode** for monitoring packet flows and dynamic statistics during simulation, and **Analyzer Mode** for post-simulation evaluation of metrics such as throughput, latency, packet loss, and connectivity. QualNet also supports importing real network topologies, traffic captures (PCAP), NetFlow statistics, and router configurations to replicate "as-is" network scenarios.

QualNet includes extensive **model libraries** covering wireless and cellular protocols (Wi-Fi, LTE, 5G), military waveforms (Link-11, Link-16), satellite and underwater communications, devices such as routers and switches, mobility patterns, terrain, and environmental effects. It also supports **Human-in-the-Loop (HITL)** interaction and external interfacing with other simulators using HLA or DIS standards.

4. NetSim

NetSim is a commercial network simulator and emulator that provides tools for designing, simulating, and analyzing various networks, including wired, wireless, and cellular networks. It allows users to create network scenarios using a graphical interface and smart libraries, analyze performance metrics, and can even emulate real-time traffic from actual devices by linking the

virtual and real worlds. Researchers can modify its C code to develop custom protocols, and its API allows for integration with other software like MATLAB and Wireshark.

Key features

- **Simulation and Emulation:** Provides both virtual simulation and emulation, which fuses the virtual with real-world traffic and devices.
- **Graphical User Interface:** Features an easy-to-use GUI for creating network topologies, placing devices, and configuring settings.
- **Protocol Support:** Includes a wide range of built-in protocols for various network types, such as Wi-Fi, LTE, 5G, MANETs, and WSNs.
- **Performance Analysis:** Offers tools to visualize packet flow and analyze performance metrics like throughput, delay, and loss.
- **Customization:** Allows for custom protocol development by modifying the source C code.
- **Extensibility:** Provides APIs for integration with other software, such as MATLAB and Wireshark.
- **Learning and Research:** Serves as a powerful tool for both academic research and for gaining practical skills in networking, with applications for testing and developing new network technologies.

5. Cisco Packet Tracer

Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems that allows users to create network topologies and imitate modern computer networks. The software allows users to simulate the configuration of Cisco routers and switches using a simulated command line interface. Packet Tracer makes use of a drag and drop user interface, allowing users to add and remove simulated network devices as they see fit. The software is mainly focused towards Cisco Networking Academy students as an educational tool for helping them learn fundamental CCNA concepts. Previously students enrolled in a CCNA Academy program could freely download and use the tool free of charge for educational use.

Overview

Packet Tracer can be run on Linux, Microsoft Windows, macOS. Apps for mobile operating systems like Android and iOS were also available. Packet Tracer allows users to create simulated network topologies by dragging and dropping routers, switches and various other types of

network devices. A physical connection between devices is represented by a 'cable' item. Packet Tracer supports an array of simulated Application Layer protocols, as well as basic routing with RIP, OSPF, EIGRP, BGP, to the extents required by the current CCNA curriculum. As of version 5.3, Packet Tracer also supports the Border Gateway Protocol.

In addition to simulating certain aspects of computer networks, Packet Tracer can also be used for collaboration. As of Packet Tracer 5.0, Packet Tracer supports a multi-user system that enables multiple users to connect multiple topologies together over a computer network. Packet Tracer also allows instructors to create activities that students have to complete. Packet Tracer is often used in educational settings as a learning aid. Cisco Systems claims that Packet Tracer is useful for network experimentation.

Role in education

Packet Tracer allows students to design complex and large networks, which is often not feasible with physical hardware, due to costs. Packet Tracer is commonly used by NetAcad students, since it is available to download after creating a free account. However, due to functional limitations, it is intended by Cisco to be used only as a learning aid, not a replacement for Cisco routers and switches. The application itself only has a small number of features found within the actual hardware running a current Cisco IOS version. Thus, Packet Tracer is unsuitable for modelling production networks. It has a limited command set, meaning it is not possible to practice all of the IOS commands that might be required. Packet Tracer can be useful for understanding abstract networking concepts, such as the Enhanced Interior Gateway Routing Protocol by animating these elements in a visual form. Packet Tracer is also useful in education by providing additional components, including an authoring system, network protocol simulation and improving knowledge an assessment system.

6. GNS3 (Graphical Network Simulator 3)

Graphical Network Simulator-3 (shortened to **GNS3**) is a network software emulator first released in 2008. It allows the combination of virtual and real devices, used to simulate complex networks. It uses Dynamips emulation software to simulate Cisco IOS.

GNS3 is used by many large companies including Exxon, Walmart, AT&T and NASA, and is also popular for preparation of network professional certification exams. GNS3 is used by hundreds of thousands of network engineers worldwide to emulate, configure, test and troubleshoot virtual and real networks. GNS3 allows you to run a small topology consisting of only a few devices on your laptop, to those that have many devices hosted on multiple servers or even hosted in the cloud.

GNS3 is open source, free software that you can download from http://gns3.com

It is actively developed and supported and has a growing community of over 800,000 members. GNS3 can help you prepare for certification exams such as the Cisco CCNA, but also help you test and verify real world deployments. Jeremy Grossman, the original developer of GNS3 originally created the software to help him study for his CCNP certifications. Because of that original work, you can today use to help you do the same without paying for expensive hardware.

GNS3 has allowed network engineers to virtualize real hardware devices for over 10 years. Originally only emulating Cisco devices using software called Dynamips, GNS3 has now evolved and supports many devices from multiple network vendors including Cisco virtual switches, Cisco ASAs, Brocade vRouters, Cumulus Linux switches, Docker instances, HPE VSRs, multiple Linux appliances and many others.

7. EVE-NG (Emulated Virtual Environment- Next Generation)

This is the Emulated Virtual Environment –Next Generation and it is a more advanced powerful network emulation platform used for building larger, professional-level virtual labs. Unlike GNS3 which mainly focuses on Cisco devices, EVE-NG supports a wide range of vendor images such as Cisco, Juniper, Palo Alto, Fortinet, Huawei and even Linux based services. This multi-vendor support makes it very useful for training in real enterprise network environments where different technologies work together.

One of the biggest advantages of EVE-NG is that it runs entirely through a web interface. This means users do not have to install heavy software on their laptops. Instead, all devices run from a single central server or virtual machine and the user manages everything from a browser. This setup makes it possible to build very large topologies sometimes with more than 50 devices without slowing down a personal computer. It is commonly used in advanced certification training such as CCNP security, CCIE and cyber security labs.

EVE-NG also allows multiple users to access the same lab environment at once which is ideal for classroom use, team projects and professional training centres. Because it is stable, scalable and flexible, EVE-NG has become one of the preferred tools for network engineers who need realistic and complex lab environments for testing, studying or demonstrating real-world scenarios.

8. Containerlab

Containerlab is an open-source, command-line interface (CLI) tool designed for simplifying the creating and management of network emulation labs using containers. Containerlab allows users to identify entire network topologies using declarative files. This method does not only improve clarity and repeatability but also aligns well with modern DevOps workflows, where automation and version control are important. With the use of a Containerlab, users will be able to install and implement complex virtual networks within seconds, making it a fast and efficient option compared to traditional hypervisor-based labs. The tool firstly focuses on running containerized network operating systems (NOS) like FRRouting and Artists among others. However, it as well brings about the flexibility to integrate traditional virtual machines (VMs), allowing users to combine containerized and virtualized network elements in the same network topology. Containerlab is often more valuable in environments that rely on network automation, testing, and continuous integration (CI) pipelines. Engineers can automatically spin up test networks, run automated scripts, validate configurations, and tear everything down while saving both time and resources. This is why Containerlab is a practical solution for developers, students, and network professionals who need an environment with scalability and efficiency and can be reproduced for experimentation or pre-deployment testing.

9. OPNET

OPNET is a network simulator that creates and analyzes communication network performance using discrete event simulations. It allows users to model various network topologies, protocols, and devices to test "what-if" scenarios, troubleshoot issues, and plan network designs. The OPNET Modeler is a powerful version that allows editing of protocol models, while the academic version is more limited with pre-built models.

Key features and uses

- **Modeling and simulation:** Create and simulate a wide range of network topologies with pre-built models for devices and protocols.
- **Performance analysis:** Evaluate network performance by analyzing data like latency, throughput, and packet loss.
- **Protocol development:** The full OPNET Modeler allows for detailed protocol modeling and modification, including editing the source code of existing protocols and creating new ones.

- **Application troubleshooting:** Simulate a network to help identify and troubleshoot issues with applications.
- **Network planning:** Assist in planning and designing networks for small to large-scale applications by validating hardware and analyzing traffic patterns before implementation.
- **Educational tool:** Used in universities to help students understand network protocols and dynamics by visualizing their behavior in a simulated environment.

Different versions

- **OPNET Modeler:** A comprehensive version that provides access to protocol models, allowing for their editing and the creation of new models.
- **OPNET IT Guru Academic Edition:** A simplified version for academic institutions that includes pre-built models but limits the ability to modify or create new protocols.

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